Arista Access Node BAS-ACC-06 Routing Protocol Flap Troubleshooting SOP

Node: BAS-ACC-06 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2098

Description

The Basavanagudi POP Arista Access Node BAS-ACC-06 is experiencing a routing protocol flap failure (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) as indicated by the ALARM_2098 alarm code. This failure domain is due to the complex interplay of routing protocols (BGP, OSPF, and ISIS) on the Arista Access Node, which is configured as an exterior gateway to the wider internet. The node's role in the network, as an aggregator of routing information and disseminator of reachability information to the internet, makes it particularly susceptible to routing protocol flaps. The Arista EOS (Extensible Operating System) software, upon which the node is based, utilizes a distributed architecture for routing protocol processing, which can lead to flapping when faced with high levels of network churn or misconfiguration. Furthermore, the lack of adequate redundancy in the link configuration may exacerbate the situation, rendering the node unable to maintain its routing protocols, leading to a cascade of failures.

Pre-requisite Checks

- Ensure that the Basavanagudi POP is accessible via remote hands.
- Verify that the redundant link is operational and providing alternative routing.
- Check the link utilization and ensure it is within acceptable limits.

Potential Root Causes

Cause	Probability
BGP route churn due to misconfigured routing policies	High
OSPF adjacency loss due to network congestion	Medium
ISIS LSP flooding due to misconfigured LDP	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 , Description: This command will verify the status of the BGP table, checking for any anomalies or inconsistencies that may indicate a routing protocol flap.

Command: show ospf neighbor, Expected Output: Address Interface State Pri Dead Time Address, Description: This command will verify the status of OSPF adjacencies on the node, checking for any issues that may indicate a routing protocol flap.

Command: show isis database, Expected Output: ISIS LSP Database: , Description: This command will verify the status of the ISIS LSP
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database on the node, checking for any issues that may indicate a
routing protocol flap.
```

Resolution Steps

Step 1: Step: Verify and correct BGP routing policies, Command: show bgp policy, Description: Verify that the BGP routing policies are correctly configured and not causing any issues with the routing protocol.

Step 2: Step: Verify and correct OSPF network configuration, Command: show ospf interface, Description: Verify that the OSPF network configuration is correct and not causing any issues with the routing protocol.

Step 3: Step: Verify and correct ISIS LDP configuration, Command: show isis ldp, Description: Verify that the ISIS LDP configuration is correct and not causing any issues with the routing protocol.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 1,
main routing table version is 1 , Description: This command will
verify that the routing protocol flap has been resolved.

Command: show ospf neighbor, Expected Output: Address Interface State
Pri Dead Time Address, Description: This command will verify that the
OSPF adjacencies are stable and not experiencing any issues.

Command: show isis database, Expected Output: ISIS LSP Database: ,
Description: This command will verify that the ISIS LSP database is
stable and not experiencing any issues.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: rollback /bootflash/configs/, Description: If the fix fails, this step will be used to restore the previous configuration and revert to the previous state.